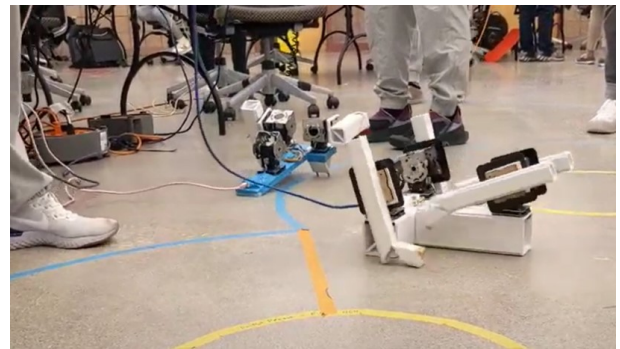


EECS 464: Hands-on-Robotics Project 0 Report

Red Team

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1 Background

1.1 Project Task Specification

As specified in the [2022F Project 0 Task Specification](#), we were to build a robot that will race in a figure 8 track on P-Day. The rule of the race was to go around the track once and come back to your starting position and not have any part of your robot crossing the outer edge of the marking tape that delineates the track. But to get a full grade for this project, we would only have to demonstrate that our robot could move from one end of the track to the other in a straight line and adhere to a set of structural requirements. The structural requirement said that no robot may have more than 3 modules (a module is a servo motor wrapped metal plates attached as shown in Figure 2) in joint configuration, be bigger than 50 by 50 by 25 cm at deployment or the track at any time, have any fully rotating body parts, have any adhesives on the modules, be made of more than 6 3D-printed parts, each no more than 3cc of filament, or made of anything but foamcore, tape, hot glue or rubber.

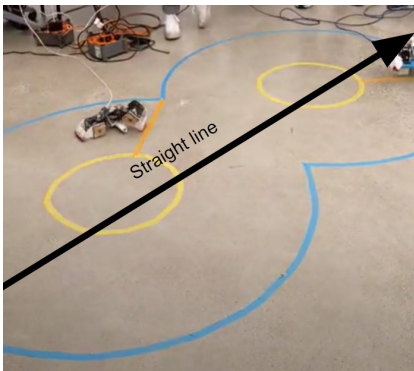


Figure 1: Figure 8 track

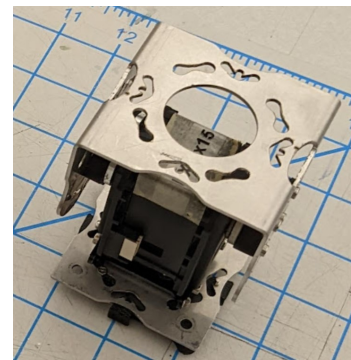


Figure 2: A module

1.2 Possible Designs

We thought of a few design, inspired by works done by the previous teams, research projects from outside entities and some just from our imagination. More details can found in [our resource document](#). A brief overview of each design is provided below.

1.2.1 Friction Bot

After reading through reports by previous teams, what we found was that the locomotion tended to be generated using some variation of the "gorilla" leg as depicted in Figure 3, whereby swinging in one direction would push the robot forward but swinging in the other direction would simply bend the leg and not move the robot. Inspired by this idea and some basic knowledge of friction, we came up with this Friction Bot (Figure 4). The front leg is simply a gorilla leg and rear two legs does nothing but adding friction to help change direction. We didn't pursue this design, because Professor Revzen said it was done by previous team.

1.2.2 Cheetah Bot

Our next idea was inspired by last year's purple team (Figure 5). Their idea was using two pairs of gorilla legs, one in the front and the other in the back for pushing and the body would twist to change direction. Adding to their idea, we decided to program our robot so that the front and back legs would alternate in pushing, like how a cheetah ran.

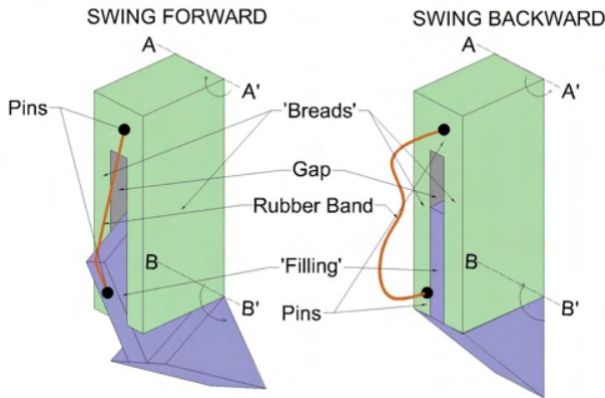


Figure 3: A type of gorilla leg [1]

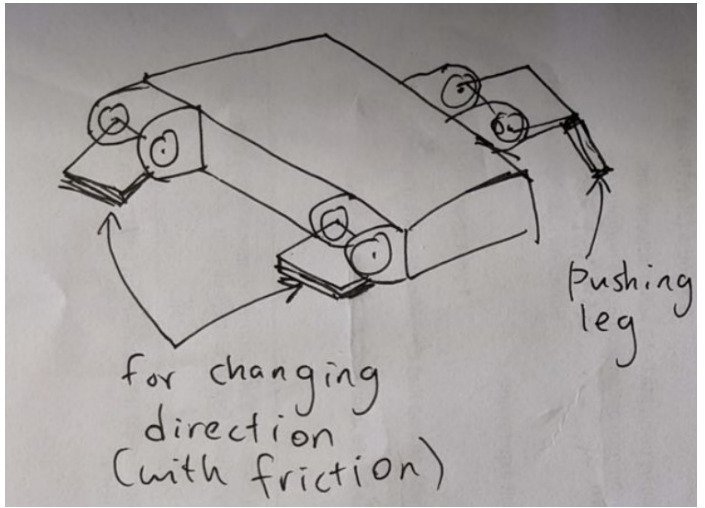


Figure 4: Friction Bot

1.2.3 Mosquito Bot

This was inspired by a robot (Figure 6) in a YouTube video [2]. The rear pair of legs stays on the ground and does nothing. The front two legs alternate in pushing, left and right. The bot turns if one of the front legs takes extra turns pushing. Although Professor Revzen encouraged us to pursue this novel design, we later found the locomotion impossible to execute with 3 modules.

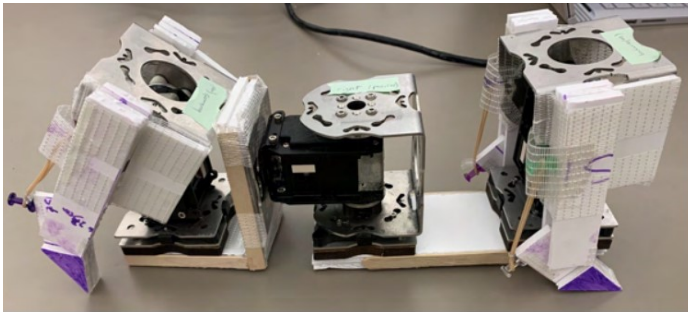


Figure 5: Last year purple team's bot [1]



Figure 6: Inspiration for Mosquito Bot [2]

1.3 Final Design

We eventually settled on a variation of the Cheetah Bot. The following two sections describe the mechanical design and software design of our group's robot. More details on how to build the robot can be found in [the how-to document](#).

1.3.1 Mechanical Design

As shown in Figure 7, our robot consisted of three main parts, a T-shaped base, three motors and four legs. The T-shaped base was designed to serve the front two and back one layout of motors. Considering the weight of the foamcore, the overall weight of the robot would increase by 30 percentage compared to the final hollow base composed of one foamcore if it were stacked by layers of foamcore to the desired base height. And we inserted two foamcore in X shape inside the base to enhance the stability and strength of the hollow base design as shown in Figure 9. Our robot used all three motors provided by the project and adopted a front-two-back layout, with one leg connected to each of the two motors on one side and one leg connected to both sides of the motors on the back. This layout not only allowed for straight-line

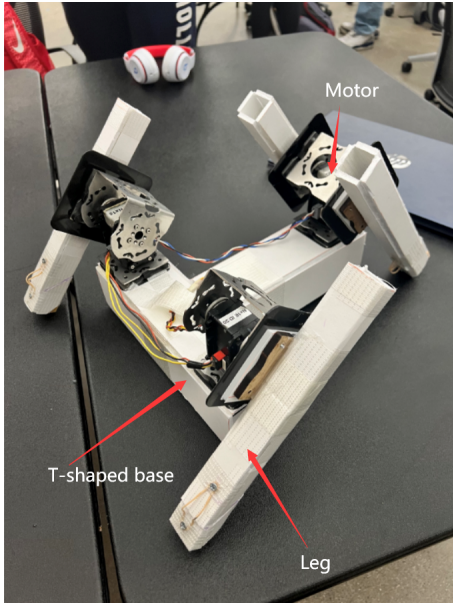


Figure 7: Red team robot

motion, such as the front and back four legs moving in the same direction at the same time or alternating back and forth, but also allowed for turning motion, such as using only the front leg to turn on one side or cooperating with the back leg to turn.

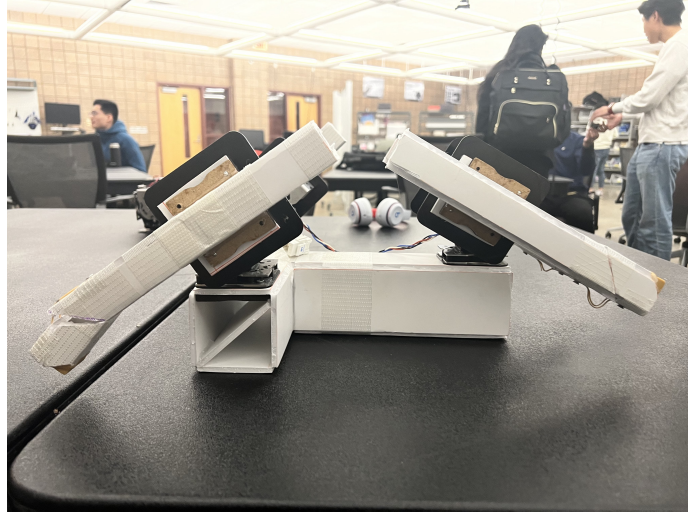


Figure 8: Side view of robot

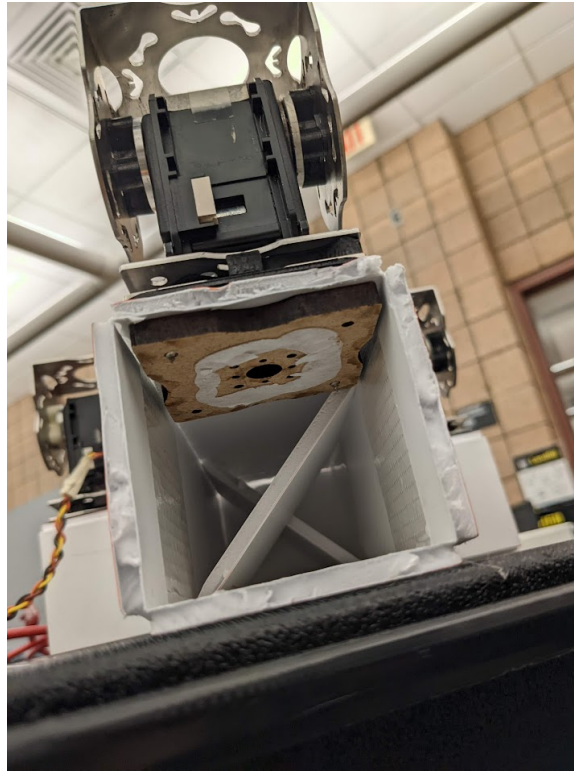


Figure 9: X foamcore support structure

1.3.2 Software Design

We mainly relied on posable programming to set robot's motion. It is an approach to quickly prototype robot motions by recording sequences of robot poses and playing them back. During setting and measuring process, we tried to make the legs lift as low as possible, which got rid of unnecessary motions.

To actually control the robot, we implemented a controller file based on `demo-shavenhaircut.py` using `joyapp` packages. Then, we put the csv files generated for corresponding key-binds. In our case, we created

following commands: two-step forward(W), one-step forward(S), turn left(A), turn right(D), turn left and forward(Q), and turn right and forward(E). We set the command to run individually, so there is no conflict between them. We made such many commands for different scenarios. Turning while forwarding would be much efficient for moving in a curve, and small steps are great for adjustment.

It is also worth notice that keyframe is actually running the motion much smoother than joyapp does. It is because the exported csv only record the position for each second. Keyframes interpolates the position every 0.03 second as we measured. To emulate the same smoothness, which we believe would help overall performance, we wrote a extension to convert the csv to smaller scale.

1	t	x	y	z
2	0.0	-4149	10000	10000
3	0.03	-3704	9574	9574
4	0.06	-3260	9148	9148
5	0.09	-2816	8722	8722
6	0.12	-2372	8296	8296

Figure 10: Portion of f.csv

2 Result

For P-Day, each team participated in a number of events. The first event was a qualifier, where each robot was timed as it walked across the length of the figure 8 track. As shown in Figure 1, the track was 6 meters by 50 cm with 2 turns <120deg with a 25cm inner radius. Then each robot was timed on its traversal through the track, where both taped boundaries counted for disqualification. During this event, only Red team and Maize team succeeded in traversing the figure 8 without stepping or reaching out of bounds. The following event was a figure 8 traversal, where the outer boundary did not count for disqualification. Every team was able to complete their journey across the track. Finally, the competition ended with a straight race. Only one team, Purple team, was unable to complete this event. The robots with the fastest times for each competition were:

- Qualification: Red team
- Strict Track Figure 8: Maize team
- Out of Bounds Released figure 8: Maize team
- Straight race: Red team

2.1 Qualification Results

The Qualifier event consisted of each robot crossing the figure 8 from end to end. Each robot's journey was timed. Those who did not cross the track in its entirety were disqualified. We had the best time of 26:94 and Purple team had the worst time of 110:76

Team	Red	Maize	Blue	Green	Purple
Time Result(seed placement)	26.94	34.72	39.90	73.26	110.76

2.2 Track Results

Team	Maize	Red	Blue	Green	Purple
Figure Eight Race	1:15	4:03.52	-	-	-
Figure Eight Race (ignore outer tape)	1:28; 1:22	1:31	4:57; 3:01	4:54	5:55

2.2.1 Strict Track Results

This competition consisted of each robot racing to complete a figure 8 track while staying between both boundaries. The only two teams to successfully complete the event were Maize team and us, Red team. The rest of the teams were disqualified for crossing either one of the boundaries.

2.2.2 Out of Bounds Released Results

This competition consisted of each robot racing to complete a figure 8 track, where the outer boundary did not count for disqualification. All teams were able to complete this competition, with Maize team having the best time of 1:22 and Purple team having the worst time of 5:55

2.3 Straight Race Results

This competition consisted of each robot racing down a straight track without overstepping the boundaries. All teams but one, Purple team, were able to complete this competition. Our robot had the best time of 37:81 and Green team had the worst time of 2:17.

Team	Red	Maize	Blue	Green	Purple
Straight Race	45.82; 37.81	1:10	1:20	2:17	-

3 Discussion

3.1 Overall Analysis

Overall, the two most consistent robots in terms of event completion times were Maize team and us, Red team. Our robot's forward movement speed and step size was bolstered by the robot's design. Each complete movement included a full forward-leg motion, followed by a complete back-leg movement. This resulted in our robot having the fastest times where direct forward movement dominated. Those events include the Qualifier and the Straight Race. Maize team's design was centered around control of angled motion. Their small size and ease of turning allowed them to outperform all teams in the figure 8 challenges. In the Strict Track, Maize team's robot beat us by 2:48:52. In the Out of Bounds Released, Maize team beat us by 9 seconds.

3.2 Our Design

The design we decided on was a modification of the gorilla structure. The front legs each were powered by their own motor; the hind legs were powered by one motor. The modification came in the way forward motion was completed. Each forward motion included a full step from our front legs which was followed by a full step from the hind legs. This essentially caused us to move twice for each forward key press. We hypothesized that having longer legs (26cm) - which raised the chassis above the ground in each forward motion - would result in farther movement. Thus, we expected farther movement to result in faster time since more ground was covered per motion. This hypothesis was supported by our winning times in each forward-motion dominant event.

Our biggest design challenge was turning efficiently. We originally had trouble making steep turns because our legs lacked a friction point, thus making it tougher to turn the large body. To ameliorate this, we added rubber pieces to the heels of each leg. This allowed each step to contribute a steeper angled motion. A common obstacle we faced while turning was our legs crossing the boundary even while the robot was in bounds. We remedied this by limiting how far in front the front legs would position themselves after each completed motion. Our idea was if they stayed as close to the ground as possible, then we could limit how often the legs crossed over the boundaries. Because of this, we believe the biggest improvement to make in the future is a size reduction of the entire robot. The structure and movement of the robot were fit to our standards. The only downfall was being disqualified because of the robot's large size.

3.3 Other Designs

3.3.1 Team Maize

Maize team's design was successful in the figure 8 competition's because of their small size and turning ability. They followed the inchworm design, where three motors were put together in a line. The front and back motors were for moving the robot, whereas the middle motor was for turning the head of the robot. Although this design allowed for efficient turning and control of motion, it lacked in speed. The small design did not allow for large steps, thus their forward motion was relatively slow.

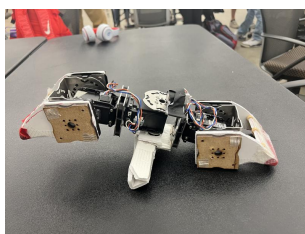


Figure 11: Maize Team's Robot

3.3.2 Team Green

Green team had a modified gorilla design that had no hind legs. They had two legs each with their own motor. The legs were placed in the middle of the body, rather than the front. A third motor was placed between the other two motors. It might have been hypothesized that 3 motors moving in unison would bolster torque. The body of the robot resembled a single rectangular shape.

The Green team's robot struggled in the different events. They ended with the second highest time in the Qualifying event and were unable to complete the Strict Track event. They completed the straight race with the slowest time. I believe a challenge they faced was a lack of control in their motions. Both forward motion and turning motion were sporadic and unpredictable. Using the Qualifier video as a reference, it is evident that their forward motion did not always stay in a straight line. A potential improvement could have been larger legs and an inclusion of a better friction point for the legs. However, I think the best improvement could be adding a set of hind legs to increase their stability.



Figure 12: Green Team's Robot

3.3.3 Team Blue

Blue team's design resembled a gorilla design without hind legs. The had two large legs in the front, each sporting their own servo motor. The body of the design was a T shape, with the bottom of the T touching the ground. A third motor was placed along the center of the body as a counter weight.

Blue team's robot had the ability to move forward, albeit at a slow pace. Their turning ability was stunted by the lack of friction at the tips of the legs. This issue of turning was evident on P-Day. The robot's forward motion was sporadic, with each step not always resulting in a straight movement. They were disqualified from the Strict Track but were able to complete Out of Bounds Released event. They scored the second highest time in that event. An improvement that could have been made to the design is changing the heel to be flat rather than pointed, and adding a rubber piece that provided more friction

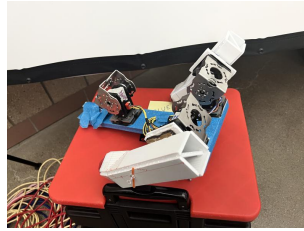


Figure 13: Blue Team's Robot

3.3.4 Team Purple

Purple Team had the largest design out of the 5 groups. It was a 2 legged Gorilla design where the body's shape was dictated mostly by its height. There was no counter balance shape for the body. The width of the robot was simply the width of the legs. The two servos were placed high above the ground at the top of the body. The center of the body helped stabilize the robot

Purple team struggled in the different events. They were unable to complete the Strict Track and Straight races. They ended both the Qualifier and Out of Bounds Released events with the slowest times. A main disadvantage that their design held was lack of stability and direction. This may be due to the height of the robot, and the lack of counter balances along the length of the robot. For example, Blue team's T shape gave them a counter balance from the rear. An improvement to make for the future would be to shrink the size of the whole robot, and add some extension of foamcore along the length to balance the robot.

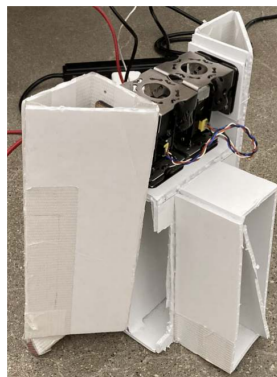


Figure 14: Purple Team's Robot

4 Conclusion

Based on the results of the competition and our observations, we believe that the main factors that determine the outcome of the races are the stability of the robot structure and the friction caused by the contact with the ground during movement. For example, Team Purple was unable to finish the race due to the instability of the robot structure during the straight line race, and Team Blue and Team Purple lost their steering ability during the race due to the reduced friction between the robot and the ground. On the other hand, our team's and Team Maize's robots were extremely stable throughout the competition, both in terms of structural stability and friction on the ground during movement, and after completing several matches the robot's straight and turning movements were barely affected by the wear and tear of the races.

5 Reference

1. [Winter 2022 Purple Team's Final Report](#)
2. [Mosquito Bot inspiration YouTube video](#)
3. [Green Team Qualifying Video](#)